

AI Safety Gateway — Product Requirements Document (PRD)

Version 1.0 • Production-grade AI Gateway for RAG Systems

Executive Summary

AI Safety Gateway is a centralized middleware that sits between applications and LLM providers. It standardizes authentication, guardrails, RAG retrieval, evaluation, observability and policy enforcement so teams do not rebuild those concerns in every AI application.

Problem Statement

Organizations often connect every AI application directly to an LLM. This duplicates authentication, rate limiting, prompt-injection protection, logging and evaluation logic while increasing security risk.

Vision

Build a Docker-first AI platform that acts like an API Gateway for LLM traffic, enforcing safety and governance while remaining model-agnostic.

Target Users

Platform engineers, AI engineers, backend engineers, security teams and internal product teams.

Success Metrics

High groundedness, low hallucination rate, low latency, zero unauthorized document exposure and continuous evaluation before releases.

Core Architecture



Every request flows through authentication, rate limiting, input guardrails, retrieval policies, generation, output verification and evaluation logging.

Functional Requirements

Capability	Description
Authentication	JWT/API Keys
Guardrails	Prompt injection, PII, toxicity, hallucination checks
RAG	Hybrid retrieval with policy filtering
Evaluation	Golden datasets, faithfulness, recall
Observability	Prometheus, Grafana, OpenTelemetry
Routing	Support multiple LLM providers

Guardrails

- Input Guardrails: detect prompt injection, jailbreaks and abusive inputs.
- Retrieval Guardrails: enforce RBAC and tenant isolation before documents reach the LLM.
- Output Guardrails: verify citations, redact PII and reject unsupported claims.

Evaluation Strategy

- Offline golden-dataset testing before deployment.
- Continuous scoring for faithfulness, context recall and answer correctness.
- Online monitoring with user feedback and production traces.

Docker-First Development

The entire platform is designed to run with Docker Compose. No local installation of PostgreSQL, Redis, Kafka, Prometheus or Grafana is required beyond Docker itself.

Repository Vision

apps/gateway apps/worker services/rag services/guardrails services/evals
infrastructure/docker k8s docs tests

Future Expansion

- Multi-tenancy with organization-level policies.
- Dynamic model routing based on cost and latency.
- Semantic caching with Redis.
- Human approval workflows.
- MCP tool integrations.
- Kubernetes deployment with autoscaling.
- Enterprise governance dashboards.

Milestone Roadmap

Phase	Goal
V1	Secure AI Gateway with RAG and guardrails
V2	Observability and automated evaluations
V3	Enterprise routing, MCP tools and multi-tenancy

Interview Value

This project demonstrates backend engineering, distributed systems, AI safety, RAG architecture, observability and production-oriented design rather than only building another chatbot.